D03 · Yield & Loss Reduction

01 · OEE — the 16-loss framework

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

The 16-loss framework decomposes total equipment effectiveness into the specific causes that degrade availability, performance, and quality — eight equipment losses (breakdown, setup, tool change, startup, minor stops, speed loss, defect, rework) and eight broader operational losses (planned downtime, management losses, motion, line organization, logistics, measurement & adjustment, energy, yield). It is the diagnostic apparatus that converts OEE from a number into an action plan.

2 WHY IT MATTERS

OEE is the most universally tracked manufacturing KPI and the most universally aggregated — and aggregation is exactly what destroys its diagnostic value. **An OEE of 65% is not a state of operation; it is an arithmetic average that conceals which 4–5 of 16 distinct losses are dominating**. The foundational text on TPM and the 16-loss decomposition makes this explicit **(Nakajima, 1988 — peer-reviewed, "Introduction to TPM", Productivity Press)**: equipment effectiveness must be decomposed into its constituent losses to be actionable, and the top three losses in any plant typically account for 60–80% of total OEE gap. The asymmetry is enormous — **a plant that attacks all 16 losses simultaneously dilutes effort and produces no measurable improvement; a plant that identifies and attacks the top three captures 60–80% of the available gain**. Most plants do the first; few do the second, because the first feels comprehensive and the second feels like it leaves things out.

3 HOW IT IS DONE

1. **Decompose OEE into its three components first — availability, performance, quality — then into the 16 sub-losses.** Most plants stop at the three-component decomposition (availability 88%, performance 79%, quality 94%, OEE 65%) and act on whichever component looks lowest. That is the wrong cut. The right cut is which of the 16 underlying losses is the largest dollar contributor, regardless of which top-level component it belongs to.
2. **Instrument each loss category with reason codes captured at the source — not reconstructed at end-of-shift.** Reason codes captured by the operator at the moment of the loss (push-button on the cell terminal, voice annotation, automatic PLC-driven categorization where possible) produce trustable data; reason codes reconstructed end-of-shift from memory produce comforting fiction. The captured-vs-reconstructed asymmetry is the single biggest determinant of whether 16-loss analysis works.
3. **Pareto the losses by dollar impact, not by minute count — and re-rank quarterly.** A 240-minute downtime event on a $200/minute line costs $48K; a series of 4,800 minor stops at 30 seconds each on the same line costs $480K. Ranking by minutes makes the downtime look bigger; ranking by dollars exposes which loss to attack first. Re-rank quarterly because the dominant losses shift as fixes land.
4. **Attack the top three losses with dedicated kaizen events — not with general improvement initiatives.** A focused 5-day kaizen on the dominant loss (cross-functional team, baseline measurement, root-cause analysis, countermeasure design, pilot, validation) consistently outperforms a 12-month broad continuous-improvement initiative on the same loss. Concentration of effort matters more than scope of intent.
5. **Validate the gain in OEE at the cell, not in aggregated reporting.** A kaizen that improves the dominant loss must be verifiable as a sustained shift in the OEE trend at the cell where the work was done — not as a plant-level number that washes out the gain in noise from other cells. Cell-level validation forces the discipline of sustained improvement; aggregated reporting forgives regression.
6. **Refresh the loss inventory annually — the dominant losses shift, and the analysis has to follow.** A loss that was #1 last year may now be #5; a previously-minor loss may have grown as throughput increased. The 16-loss inventory is not a one-time exercise — it is the operational diagnostic that has to be re-run yearly to keep the kaizen pipeline pointed at the real opportunity.

WORKED EXAMPLE — ILLUSTRATIVE

An automotive Tier 2 plant runs 9 production lines with plant-level OEE = 68% reported monthly. Baseline 16-loss decomposition reveals the structure that the headline number was hiding: **top 3 losses account for 71% of the 32-point OEE gap** — minor stops (14.2 OEE points lost, ~$3.1M/year), changeover time (10.8 points, ~$2.4M/year), and breakdown downtime (8.4 points, ~$1.8M/year). Losses 4–16 collectively account for the remaining 29%. The implementation: 3 sequential 5-day kaizen events over 4 months, one per dominant loss, each with cross-functional team and verified countermeasures (minor stops attacked with vision-system inspection at the choke point; changeover attacked with SMED on the highest-frequency setup; breakdown attacked with criticality-based predictive maintenance on 2 critical pumps). Results at 9 months: **OEE rises from 68% to 79%, top 3 losses reduced by 64%, plant throughput +16% with no capex on additional capacity**. The fourth-ranked loss (planned changeover for sanitation) now becomes the dominant target for the next kaizen cycle.

IN PRACTICE

Pareto by dollar impact, not by minute count

Minute-count Pareto over-weights minor stops and under-weights setup. Dollar-impact Pareto converts losses to common economic units — and almost always reveals that the top three losses by dollars are not the top three by minutes. The kaizen pipeline should follow the dollars.

Attack the top three losses with dedicated kaizens, not broad initiatives

Top-3 losses typically account for 60–80% of the OEE gap **(Nakajima, 1988 — peer-reviewed, Productivity Press)**. Attacking all 16 simultaneously dilutes effort to zero measurable impact; attacking the top three captures the gain.

Reject kaizen tourism — charter against ranked losses

Publish the dollarized loss ranking and require every kaizen charter to name the loss-rank it attacks. Kaizens against rank 1–3 proceed; against rank 4–16 require explicit justification. Bureaucratic and necessary — without it the program drifts toward popularity rather than economics.

Refresh the loss inventory annually as equipment and product mix drift

An OEE program with a three-year-old loss inventory is attacking yesterday's losses. Equipment wears, products reformulate, demand mix shifts — the top three losses move. The annual refresh is what keeps the program aligned to current economics.

WHAT YOU WOULD USE

OEE & 16-LOSS DECOMPOSITION

MES-native OEE with loss reason codes (Siemens Opcenter, Rockwell FactoryTalk, AVEVA MES) — When the MES is in place, native OEE modules support 16-loss reason coding with operator capture — depth of decomposition depends on configuration discipline, not on the platform **(Siemens — vendor; Rockwell — vendor; AVEVA — vendor)**.

MachineMetrics / Memex / Augury (production analytics tier) — Specialized OEE platforms that deploy faster than full MES, with native loss decomposition, predictive analytics, and dollarized ranking dashboards; strong fit for plants that want 16-loss analysis before committing to full MES **(MachineMetrics — vendor; Augury — vendor)**.

Power BI / Tableau dashboards over MES + ERP cost data — The visualization layer that joins MES loss-count data with ERP cost-per-minute data to produce true dollarized loss ranking; quality is determined by the data engineering, not by the tool **(Microsoft — vendor; Salesforce/Tableau — vendor)**.

PLC-direct OEE counter + a disciplined operator-driven reason code log — For lean operations without MES — direct readout of run/stop status from the PLC, paired with a structured paper or tablet log of reason codes captured by the operator at the moment of the stop. Works under TPM if discipline holds; the bridge solution, not the destination.

4 SO WHAT

OEE without the 16-loss decomposition is a number, not a diagnosis — and numbers do not improve, diagnoses do. Decompose into the three components first then the 16 sub-losses, instrument with reason codes captured at the source not reconstructed end-of-shift, Pareto by dollar impact not by minute count, attack the top three losses with dedicated kaizens not broad initiatives, validate gain at the cell not in aggregated reporting, and refresh the inventory annually. The framework is 60 years old; the difference between plants that improve OEE and plants that report it is whether the decomposition is the operational state or the quarterly homework.